

Ziad Mahmoud

Software Tester & Full-Stack Web Developer

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SUMMARY

Software Tester and Full-Stack Web Developer, graduate of the Faculty of Computers and Artificial Intelligence at Helwan University. Hands-on experience in software testing, test automation, and full-stack web development gained through academic and graduation projects, with additional experience in data engineering, business intelligence, and virtualization infrastructure. Currently preparing for the ISTQB CTFL certification, while actively expanding into AI/LLM tooling, cybersecurity fundamentals, and innovation and entrepreneurship programs through continuous self-directed learning.

TECHNICAL SKILLS

Software Testing: Manual Testing, Test Case Design, Bug Reporting, Functional Testing, Regression Testing, API Testing, QA, Test Automation

Web Development: Frontend Development, Backend Development, Full-Stack Development, REST APIs, React, Django, HTML, CSS, JavaScript

Programming Languages: Python, JavaScript, C++, .NET

Databases & Tools: Databases, Database Integration, Git, GitHub, Deployment

AI & Data: Artificial Intelligence, Machine Learning, Data Analysis, Keras, TensorFlow, Streamlit

Data Engineering & BI: Power BI, SSIS, SSDT, DAX, Power Query, dbt, Apache Airflow, Apache NiFi, Docker

Infrastructure & Virtualization: VMware vSphere, ESXi, vCenter, Ubuntu Server

EXPERIENCE

Full-Stack Web Development | Academic & Graduation Projects

- Built and shipped complete web applications end-to-end — frontend interfaces, backend logic, databases, and deployment — using React, JavaScript, Django, and .NET.
- Applied structured testing (Cypress, manual QA) throughout the development lifecycle to catch issues before release.

Tools: React, JavaScript, Django, .NET, REST APIs, Databases, Cypress

Data Engineering, BI & Machine Learning | Academic & Team Projects

- Built ETL/data pipelines (SSIS, Apache NiFi, dbt) and Power BI dashboards with custom DAX measures to support business decision-making.
- Trained and compared machine learning models for a real-world classification problem, deploying the result as an interactive app.

Tools: SSIS, Apache NiFi, dbt, Apache Airflow, Power BI, Python, Scikit-learn, TensorFlow, Streamlit

Infrastructure & Virtualization | Academic Project

- Managed a multi-host VMware ESXi cluster through vCenter, including live VM migration (vMotion), snapshotting, and OVF-based backup workflows.
- Tested API connectivity across virtual machines in a simulated production environment.

Tools: VMware vSphere, ESXi, vCenter, Ubuntu Server

Team Leadership & Innovation | InnovEgypt Program — ITIDA & TIEC, 45-Hour Innovation Training

- Rotated through Tech, Team Lead, and Operations roles across a 45-hour innovation program, then collaborated with a cross-functional team to ideate a startup concept: a personal finance app helping Gen Z track spending and plan salary allocation.
- Presented the final project for evaluation, earning a Certificate of Acknowledgement from ITIDA & TIEC.

Tools: Team Leadership, Product Ideation, Operations Coordination, Startup Thinking

PROJECTS

Pharmacy Management System | Graduation Project

A web-based pharmacy management system with a React frontend, API-integrated pharmacy and product data, and Cypress-tested workflows.

Tools: React, JavaScript, Tailwind CSS, Cypress

MCS Offshore Evaluation System | Full-Stack Application

A full-stack employee evaluation platform with authentication, evaluation workflows, and QA-tested deployment.

Tools: Full-Stack Development, Database Integration, QA & Testing

Rayyes | Restaurant QR Ordering System

A multi-tenant QR ordering system where customers scan a code to browse the menu and order, with roles/permissions and tenant isolation built in.

Tools: Backend Planning, Roles & Permissions, System Architecture

Credit Risk Classification | Data Science Project

A comparative machine learning project predicting loan default risk, reaching 93.46% test accuracy with Random Forest and deployed as an interactive Streamlit app.

Tools: Python, Scikit-learn, Keras, TensorFlow, Streamlit

Insurance Claims Decision Support System | Team Project, 6 members

An ETL and BI system unifying claims data from CSV, TXT, and Excel sources into a Power BI dashboard with custom DAX measures.

Tools: SSIS, SSDT, SQL Server, Power BI, DAX

Agriculture Data Warehouse Pipeline | Data Engineering Project

A production-style pipeline ingesting agricultural yield data via Apache NiFi, transforming it with dbt, and orchestrating it with Airflow into a Power BI-ready analytics layer.

Tools: Apache NiFi, dbt, Apache Airflow, Docker, Power BI

VMware vSphere ICM Project | Infrastructure / Virtualization

A two-host ESXi cluster managed through vCenter, testing live VM migration, snapshots, and cross-host API connectivity.

Tools: VMware vSphere, ESXi, vCenter, Ubuntu Server, Node.js

EDUCATION

Bachelor's Degree, Faculty of Computers and Artificial Intelligence | Helwan University

2026

CERTIFICATIONS

- ISTQB Certified Tester Foundation Level (CTFL) — Currently Preparing
- Introduction to Cybersecurity — Cisco Networking Academy, via E-Learning Competence Center (ELCC) (March 2026)
- Introduction to Generative AI Studio — Google Cloud, via Simplilearn SkillUp (March 2026)
- Certificate of Acknowledgement, InnovEgypt Program — ITIDA & TIEC (45-hour training course and startup project)
- Build with AI: Masr Edition — Certificate of Attendance, Google for Developers & ITI
- Claude 101 — Certificate of Completion, Anthropic